

MATH 4360 HOMEWORK ASSIGNMENT 6

DUE ON THURSDAY 12 OCTOBER 2022

Please turn in your homework in class or to Weeks Hall 347 no later than 5 pm.

- (1)
 - (a) Show that $\mathbb{Z}[i\sqrt{3}]$ is not a UFD.
 - (b) Decide if 3 is a prime in $\mathbb{Z}[i\sqrt{3}]$.

- (2)
 - (a) Show that 1 is a greatest common divisor of 2 and $1 + i\sqrt{5}$ in $\mathbb{Z}[i\sqrt{5}]$.
 - (b) Show that 1 is not contained in the $\langle 2, 1 + i\sqrt{5} \rangle$ of $\mathbb{Z}[i\sqrt{5}]$.

- (3) Let G be a finite group in which every element has order at most 2.
 - (a) Show that G is abelian.
 - (b) Show that G is a $\mathbb{Z}_2$ -vector space.
 - (c) Show that the order of G is 2^n for some $n \in \mathbb{N}$.

- (4) Prove that there are infinitely many primes that can not be written as a sum of squares.